

Randomized Subspace QAOA: Certified Active-Subspace Optimization for Multi-Angle QAOA

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Abstract

Multi-angle QAOA (ma-QAOA) assigns an independent parameter to every mixer and cost term, increasing the depth- p parameter dimension to $d = p(|E| + |V|)$. We introduce Randomized Subspace QAOA (RSQ), which optimizes inside a matrix-free active subspace of the Jacobian of the per-edge observable vector. RSQ accesses the Jacobian only through Jacobian–vector and vector–Jacobian products, selects rank with a randomized residual certificate, checks that certificate during optimization, and updates a recycled basis by augmenting and recompressing it. A separate forward-only variant replaces every reverse product, including the refresh certificate, by finite-difference directional probes. We prove that the scalar ma-QAOA gradient lies in the Jacobian row space and bound the first-order objective change discarded by a certified subspace. On a released CPU benchmark of 36 paired MaxCut instances (three graph families, $n \in \{8, 10\}$, $p \in \{1, 2\}$, and three seeds), two-sided RSQ at tolerance 10^{-2} reduces the optimized dimension by 41.5% at $p = 1$ while changing approximation ratio by -0.21 ± 0.89 percentage points (mean $\pm$ two standard errors) relative to full ma-QAOA. At $p = 2$ it reduces the dimension by 70.8% with a -2.86 ± 1.43 point change. The forward-only variant is more compressed but loses more quality. Instrumented operator counts also show that subspace discovery has substantial setup cost on these small simulators; we therefore claim controlled parameter compression, not a quantum query advantage. The results identify both a useful shallow-depth regime and a failure boundary that any larger-scale RSQ claim must cross.

1 Introduction

The Quantum Approximate Optimization Algorithm (QAOA) alternates a cost Hamiltonian and a mixer, then adjusts their angles with a classical optimizer [4, 5, 15]. Multi-angle QAOA (ma-QAOA) replaces the two shared angles in each layer by one phase parameter per cost term and one mixer parameter per qubit [8]. This removes a strong tying constraint, but it changes the classical search dimension from $2p$ to

$$d = p(|E| + |V|). \quad (1)$$

The added coordinates are useful only when they create independent changes in measurable observables. Searching all of them indiscriminately can therefore spend optimization effort in directions to which the circuit is locally insensitive.

Existing reductions approach this proliferation from three directions. Symmetry-based reductions tie angles that graph automorphisms make equivalent [13]; they are exact but compress only as far as the automorphism group allows, and many random graphs have trivial symmetry. Measurement-compression and landscape-reconstruction methods exploit low-rank structure in what is measured or

in the landscape itself [7, 10], but they do not by themselves produce a reduced *search space* equipped with a quantitative acceptance test. Active-subspace methods [2, 3] seek the directions along which a function is locally most sensitive and optimize only there; they are the natural template for ma-QAOA, whose added coordinates are useful exactly when they move measurable observables. The obstacle is practical. Explicitly forming the Jacobian that defines the active subspace costs more than the search it is meant to save, and a subspace fixed once at initialization has no way to notice that it has gone stale as the optimizer moves. What is missing is a construction that touches the Jacobian only through operator actions, reports a computable certificate of the sensitivity it discards, and refreshes itself when that certificate fails. RSQ is that construction.

RSQ combines an observable-space formulation with adaptive matrix-free randomized linear algebra. The numerical primitive is fixed-tolerance randomized QB, which approximates a linear operator through products, estimates its residual from random probes, and chooses its output rank adaptively [6, 9, 14]. Our contribution is not a new general-purpose QB factorization. It is the quantum-optimization construction around that primitive: the operator is a ma-QAOA sensitivity map; its right singular space becomes the feasible search space; its estimated residual becomes a refresh criterion; and a forward-only diagnostic makes the restricted variant operational without an adjoint. The experiments then count the setup cost that this construction introduces instead of treating dimension reduction as a proxy for resource savings.

The central empirical question is not whether a subspace can be constructed, but whether the resulting compression preserves optimization quality after its setup cost is counted. We answer that question on a deliberately small, fully released benchmark. The answer is conditional: at depth one, RSQ closely tracks full ma-QAOA while optimizing substantially fewer coordinates; at depth two, stronger compression incurs a measurable quality loss; and on the studied simulator the operator setup is more expensive than direct optimization. This claim boundary is a result, because it separates structural compression from an unsubstantiated hardware-speedup claim.

Contributions.

1. We formulate the per-edge observable Jacobian as a matrix-free operator and prove that its row space contains the local gradient of any weighted MaxCut objective.
2. We give a certificate-gated optimizer with a recycled update that can rotate a saturated basis: old directions are augmented by current residual directions and the combined space is recompressed.
3. We give a genuinely forward-only build and refresh path. A dedicated test requires zero vector-Jacobian products throughout this path.
4. We release paired results, operator counters, tests, figures, and the script that regenerates every aggregate in this article.

2 Observable-space formulation

Let $G = (V, E)$ be an unweighted graph and let $C_e = (I - Z_i Z_j)/2$ for $e = (i, j)$. A depth- p ma-QAOA state is

$$|\psi(\boldsymbol{\theta})\rangle = \prod_{\ell=1}^p \left[\prod_{i \in V} e^{-i\beta_{\ell i} X_i} \prod_{e \in E} e^{-i\gamma_{\ell e} C_e} \right] |+\rangle^{\otimes |V|}. \quad (2)$$

Define the vector of local observables

$$F(\boldsymbol{\theta}) = (\langle \psi(\boldsymbol{\theta}) | C_e | \psi(\boldsymbol{\theta}) \rangle)_{e \in E} \in \mathbb{R}^m, \quad m = |E|, \quad (3)$$

and its Jacobian

$$\mathbf{J}(\boldsymbol{\theta}) = \frac{\partial F}{\partial \boldsymbol{\theta}} \in \mathbb{R}^{m \times d}. \quad (4)$$

For edge weights $w \in \mathbb{R}^m$, the expected cut is $C(\boldsymbol{\theta}) = w^\top F(\boldsymbol{\theta})$.

Proposition 1 (The local objective gradient is active). *At every differentiable point,*

$$\nabla C(\boldsymbol{\theta}) = \mathbf{J}(\boldsymbol{\theta})^\top w \in \text{range}(\mathbf{J}(\boldsymbol{\theta})^\top). \quad (5)$$

Consequently, the steepest first-order change of any weighted sum of the chosen observables lies in the parameter-space row span of $\mathbf{J}$.

Proof. Apply the chain rule to $C = w^\top F$. □

This statement is local. The row space may rotate as optimization moves, which is why a fixed subspace can fail even if it is exact at its construction point. It also gives an algebraic ceiling

$$\text{rank}(\mathbf{J}) \leq \min(m, d). \quad (6)$$

For the observable choice used here, m does not grow with QAOA depth while $d = p(m+n)$ does. Depth can therefore increase the compressible complement even without additional numerical singular-value decay.

3 Randomized Subspace QAOA

3.1 Matrix-free sensitivity actions

At an anchor $\boldsymbol{\theta}_0$, RSQ exposes two actions without materializing $\mathbf{J}$:

$$\text{jvp}(v) = \mathbf{J}(\boldsymbol{\theta}_0)v, \quad (7)$$

$$\text{vjp}(u) = \mathbf{J}(\boldsymbol{\theta}_0)^\top u. \quad (8)$$

The implementation evaluates the VJP by reverse-mode automatic differentiation. It evaluates a JVP either by automatic differentiation or by the central finite difference

$$\mathbf{J}v = \frac{F(\boldsymbol{\theta}_0 + hv) - F(\boldsymbol{\theta}_0 - hv)}{2h} + O(h^2), \quad (9)$$

with $h = 10^{-4}$. The released tests compare both actions to a dense Jacobian and verify the adjoint identity.

3.2 Adaptive basis and residual certificate

RSQ factors $A = \mathbf{J}^\top$ as $A \approx \mathbf{Q}B$, where $\mathbf{Q} \in \mathbb{R}^{d \times r}$ has orthonormal columns and $B = \mathbf{Q}^\top A$. Randomized QB expands $\mathbf{Q}$ in blocks until an estimated relative residual falls below ε . We expose a Frobenius certificate and a power-iteration spectral certificate, following established randomized matrix-free approximation [6, 11, 14]. The optimization variables are $z \in \mathbb{R}^r$ with

$$\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \mathbf{Q}z. \quad (10)$$

Every $K = 25$ steps, RSQ evaluates the residual at the current point. If the relative residual exceeds $\varepsilon_{\text{refresh}} = 0.05$, it reanchors. A naive recycled implementation can freeze: if the old basis already has the maximum requested rank, no new column can enter even when the residual is large. Our update instead permits a temporary augmented basis, adds current residual directions, and takes the singular value decomposition of the small coefficient matrix B . Rotating by its leading left singular vectors returns the best requested-rank basis available in the augmented space. This step is both implemented and regression tested.

3.3 Forward-only build and refresh

The forward-only variant selects an r -dimensional trial space by applying JVPs to a Gaussian parameter sketch and solving the resulting small Rayleigh–Ritz problem. Its refresh certificate also uses only JVPs. For independent $v_i \sim \mathcal{N}(0, I_d)$, define

$$\hat{\rho}_F^2 = \frac{\sum_{i=1}^s \|\mathbf{J}(\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)v_i\|^2}{\sum_{i=1}^s \|\mathbf{J}v_i\|^2}. \quad (11)$$

Both numerator terms are evaluated with forward differences. The identity $\mathbb{E} \|Mv\|^2 = \|M\|_F^2$ implies that the numerator and denominator converge to the corresponding squared Frobenius norms as s grows. We use $s = 12$. Unlike a spectral certificate, Eq. (11) controls average discarded sensitivity; a missed worst direction remains a failure mode.

Proposition 2 (Moment identity for the forward residual diagnostic). *Let $R = \mathbf{J}(\mathbf{I} - P)$ and let $v_1, \dots, v_s$ be independent standard Gaussian vectors. For*

$$\hat{a}_s = \frac{1}{s} \sum_{i=1}^s \|Rv_i\|^2, \quad (12)$$

we have

$$\mathbb{E}[\hat{a}_s] = \|R\|_F^2, \quad \text{Var}(\hat{a}_s) = \frac{2}{s} \left\| R^\top R \right\|_F^2. \quad (13)$$

The analogous identities hold for the denominator of Eq. (11) with R replaced by $\mathbf{J}$.

Proof. Write $\|Rv_i\|^2 = v_i^\top R^\top R v_i$ and apply the first two moments of a real Gaussian quadratic form. Averaging independent probes divides the variance by s . $\square$

Proposition 2 justifies the diagnostic as a consistent randomized estimator. It does not make the ratio of its numerator and denominator unbiased, nor does a finite probe count provide a deterministic upper bound. Accordingly, the implementation uses the observed ratio as a refresh trigger, while Theorem 1 below is conditional on the true operator residual. This distinction is important when interpreting the word “certified” in a finite randomized run.

3.4 First-order truncation bound

Theorem 1 (Certified local objective loss). *Let $P = \mathbf{Q}\mathbf{Q}^\top$ and suppose $\|\mathbf{J}(\boldsymbol{\theta})(I - P)\|_2 \leq \varepsilon$. If F is twice continuously differentiable in a neighborhood of $\boldsymbol{\theta}$, then for a step Δ ,*

$$|C(\boldsymbol{\theta} + \Delta) - C(\boldsymbol{\theta} + P\Delta)| \leq \|w\| \varepsilon \|\Delta\| + O(\|\Delta\|^2). \quad (14)$$

Proof. Taylor expansion gives $F(\boldsymbol{\theta} + \Delta) - F(\boldsymbol{\theta} + P\Delta) = \mathbf{J}(\boldsymbol{\theta})(I - P)\Delta + O(\|\Delta\|^2)$. Apply the operator-norm bound and then Cauchy–Schwarz to $w^\top [F(\boldsymbol{\theta} + \Delta) - F(\boldsymbol{\theta} + P\Delta)]$. $\square$

The theorem does not give global convergence or an approximation-ratio guarantee. It states exactly what the residual certificate controls: the first-order change discarded at the current anchor.

Algorithm. The implemented loop is:

1. construct $\mathbf{Q}$ at $\boldsymbol{\theta}_0$ to tolerance ε ;
2. run one Adam step in z for the objective $-C(\boldsymbol{\theta}_0 + \mathbf{Q}z)$;

3. every K steps, evaluate the current residual;
4. when it exceeds threshold, reanchor, augment and recompress the recycled basis, reset z , and continue.

All forward evaluations, JVPs, VJPs, ranks, and refreshes are instrumented.

4 Experimental protocol

We test unweighted MaxCut on three graph families: random 3-regular graphs, Erdos–Renyi graphs with edge probability 0.5, and rings. The grid contains $n \in \{8, 10\}$, $p \in \{1, 2\}$, and seeds $\{0, 1, 2\}$: 36 paired instances. The simulator is a pure-PyTorch, complex-128 statevector implementation [12]; an optional PennyLane backend supplies an independent cross-check [1]. Brute-force MaxCut gives the exact denominator for the approximation ratio.

Every optimizer receives 100 Adam steps. Full ma-QAOA, fixed-rank RSQ, and two-sided RSQ share the same instance-specific initial angle vector. The symmetry baseline uses the same seed in its reduced parameterization. We compare:

- full ma-QAOA;
- angles tied by graph-automorphism orbits;
- a rank-8 dense-SVD oracle fixed at the initial Jacobian (its Jacobian setup cost is excluded, so this is an oracle rather than a deployable baseline);
- two-sided adaptive RSQ at $\varepsilon \in \{10^{-1}, 10^{-2}\}$;
- forward-only rank-8 RSQ.

The primary outcome is expected-cut approximation ratio. Parameter count is the dimension optimized after construction. Paired differences subtract the full ma-QAOA ratio on the same graph and seed. We report mean and standard deviation for raw ratios and mean $\pm$ two standard errors for paired changes. No hypothesis test is used to turn absence of significance into equivalence.

5 Results

5.1 Depth one: compression with a small paired change

At $p = 1$, full ma-QAOA optimizes 22.4 parameters on average and attains 0.851 ± 0.053 . RSQ at $\varepsilon = 10^{-2}$ optimizes 13.4 parameters, a 41.5% reduction, and attains 0.849 ± 0.053 . The paired change is -0.21 ± 0.89 percentage points. This interval includes zero, but the experiment was not designed as a formal equivalence trial; the correct interpretation is that no material average gap is resolved at the precision of this released grid.

The rank-8 fixed-subspace oracle reaches only 0.756 ± 0.047 , a paired change of -9.54 ± 2.18 points. Forward-only rank-8 RSQ reaches 0.819 ± 0.049 , improving substantially on the fixed oracle but remaining 3.18 ± 1.43 points below full ma-QAOA. Adaptive refresh is therefore useful, but forward-only access does not close the depth-one quality gap at rank 8.

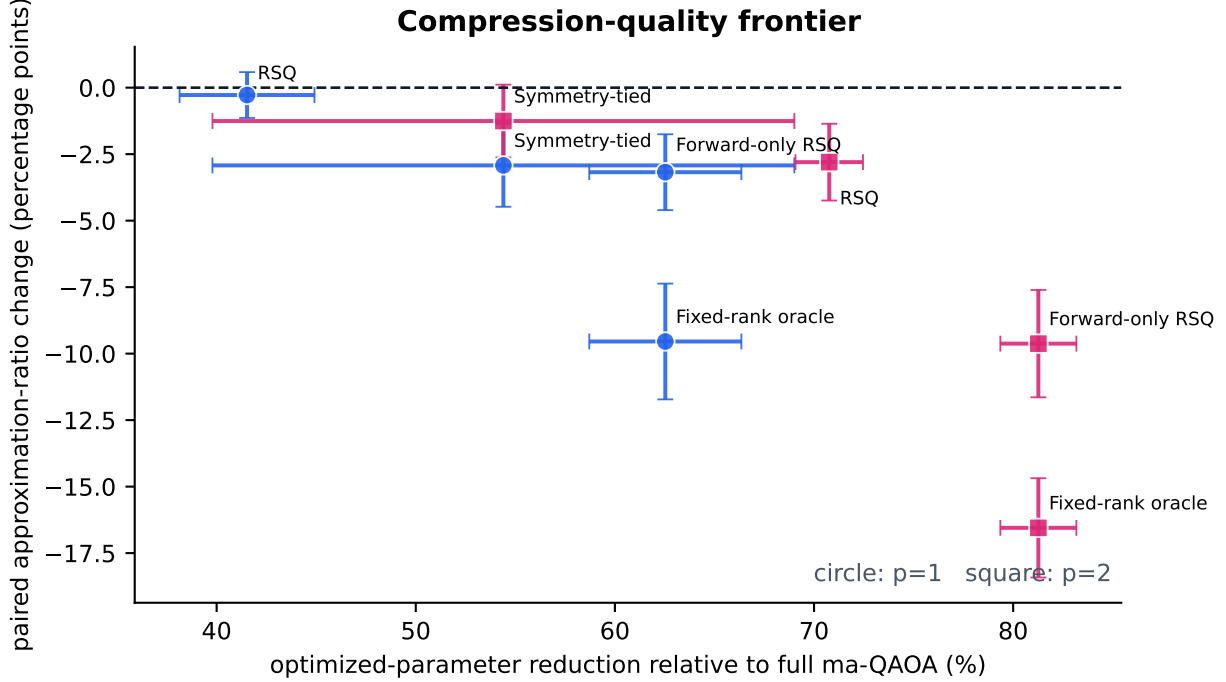


Figure 1: Compression-quality frontier. Horizontal coordinates are optimized parameter reduction relative to full ma-QAOA. Vertical coordinates are paired approximation-ratio changes; error bars show two standard errors. The main RSQ point at $p = 1$ approaches the zero-change line, whereas stronger compression and depth two expose a larger quality cost.

5.2 Depth two: the failure boundary becomes visible

At $p = 2$, full ma-QAOA optimizes 44.9 parameters on average and attains 0.925 ± 0.057 . RSQ optimizes the same 13.4-dimensional observable-controlled subspace as at depth one, now a 70.8% reduction. At tolerance 10^{-2} its ratio is 0.896 ± 0.060 , a paired change of -2.86 ± 1.43 points. The added layer grows the ambient parameter space without growing the observable count, but the more aggressive compression no longer preserves the full optimizer’s quality.

The forward-only rank-8 variant loses 9.62 ± 2.02 points at depth two. The fixed rank-8 oracle loses 16.55 ± 1.87 points. These results rule out a blanket claim that a very small initial sensitivity space is sufficient for ma-QAOA. The subspace must be large enough, must be refreshed, and may need richer observables at higher depth.

5.3 The certificate identifies algebraic, not extra numerical, rank

Across the tested tolerances, two-sided RSQ retains 13.4 directions on average, the average number of edge observables. Tightening the tolerance from 0.1 to 0.01 does not materially change the rank or the quality. Thus this benchmark does not show additional numerical low-rank decay beyond the algebraic ceiling in Eq. (6). What grows with depth is the complement between m and d , not a newly discovered decay of the nonzero spectrum.

5.4 Setup cost prevents a query-efficiency claim

At tolerance 0.1, the two-sided method averages 237 forward- F evaluations, 119 JVPs, and 358 VJPs at $p = 1$, compared with 100 forward evaluations and 100 VJPs for the full baseline under the simulator’s

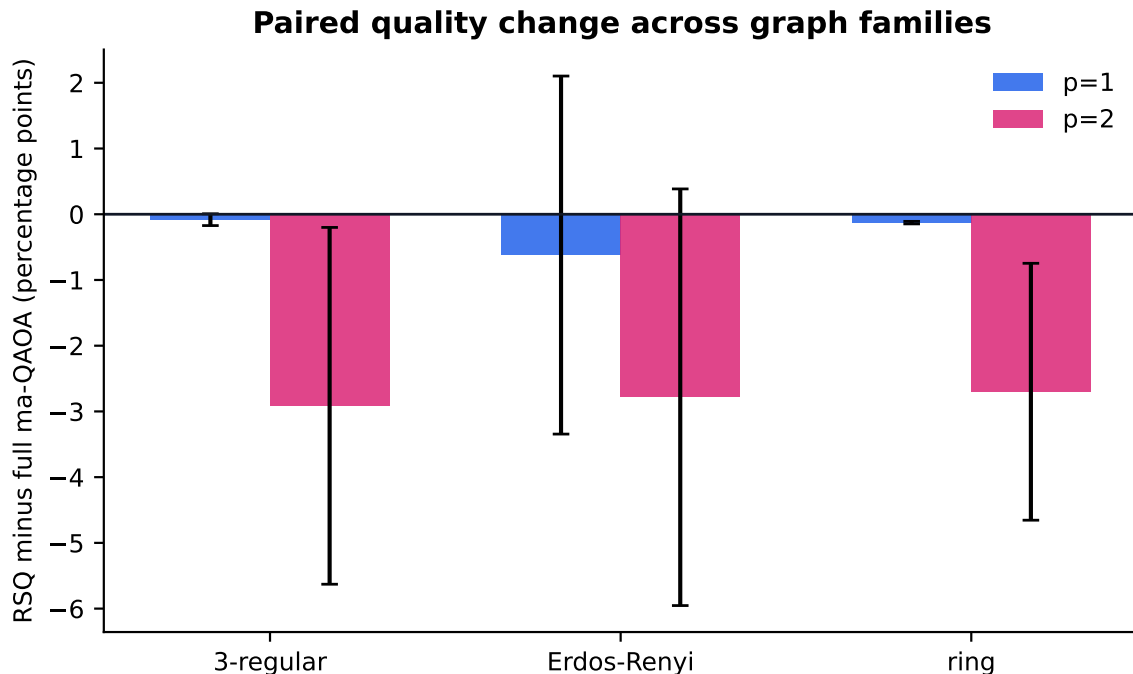


Figure 2: Paired quality change for RSQ at $\varepsilon = 0.1$, separated by graph family. Bars show means and error bars show two standard errors. The depth-two loss is present across all three tested families; wide intervals on some families reflect the small released grid.

accounting. At $p = 2$ the corresponding RSQ counts increase to 267, 134, and 408. The forward-only path uses no VJP, as intended, but requires about 285–288 forward evaluations and 143–144 JVPs. These categories are instrumentation counters, not additive physical-shot counts; a JVP by central difference contains forward evaluations already recorded by the lower-level counter.

6 Discussion

The experiments support a narrower and more useful conclusion than the original resource-efficiency hypothesis. At depth one, the observable row space removes roughly two fifths of the optimized coordinates with little resolved change in mean quality. At depth two, the same observable dimension removes roughly seven tenths of the coordinates but loses about three approximation-ratio points. Parameter compression is therefore real, while quality preservation is depth-dependent.

The comparison to the fixed-rank oracle clarifies why refresh matters. Even an initial dense SVD is poor when frozen. The relevant object is a moving local row space, not a universal low-dimensional parameterization. The repaired recycled update is essential here: simply seeding a saturated QB factorization with the old basis prevents rotation and produced a large artificial quality gap in an earlier implementation. The regression suite now checks the corrected update.

The forward-only path establishes feasibility but not parity. It avoids every VJP, including during refresh, and outperforms the equally sized fixed oracle. Its remaining quality loss can arise from restricting the Rayleigh–Ritz search to a random trial space, finite-difference error, and the use of an average Frobenius drift certificate. Hardware studies should compare larger sketches, common-random-number shot estimators, and worst-direction forward certificates.

Most importantly, fewer optimized coordinates do not automatically mean fewer quantum queries. The released small-instance counts go in the opposite direction because basis construction is expensive.

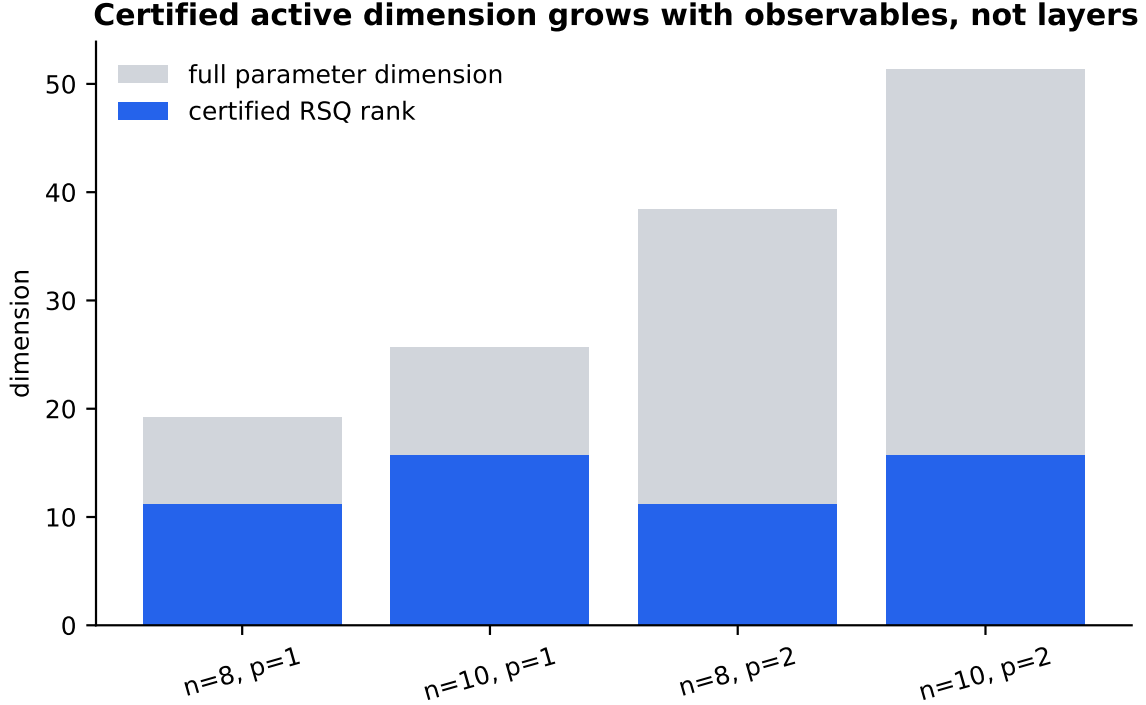


Figure 3: Ambient ma-QAOA dimension and certified RSQ rank, averaged across graph families and seeds. The observable count controls the retained dimension while the ambient parameter count doubles with depth.

A genuine query advantage would require amortizing a basis across related instances, using hardware-native batching, or reaching a scale at which full-gradient cost grows faster than the matrix-free sketch. Those are testable future conditions, not conclusions of the present data.

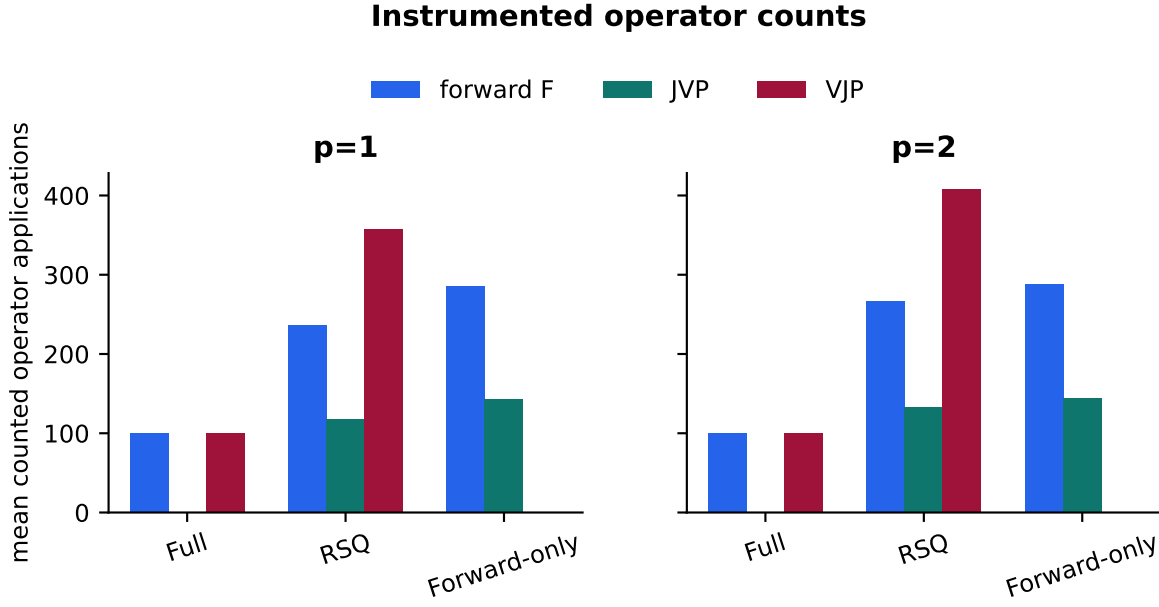
7 Limitations

The benchmark contains only 36 paired instances, $n \leq 10$, depths one and two, and exact statevector objectives. It does not model shot noise, device noise, compilation, queue latency, or weighted spin glasses. The graph families are standard but narrow. One hundred optimizer steps may favor some parameterizations over others, and no hyperparameter sweep is used to optimize each baseline separately. The fixed-rank oracle excludes its dense-Jacobian setup cost and is therefore only a quality reference. The randomized residual ratios are finite- probe estimators and can underestimate the true residual; no deterministic or run-wise coverage guarantee is claimed for the chosen probe count. Finally, the observable vector contains per-edge costs; adding mixer observables, commutators, or layer-resolved features could raise rank but improve high-depth fidelity.

8 Reproducibility

The canonical package is the repository’s `code/` directory. A clean environment can install and verify it with

```
python -m pip install -e 'code[dev,experiments]'
python -m pytest code/tests -q
```

Categories are reported separately and are not additive.

Figure 4: Mean instrumented operator counts. Categories are shown separately and are not additive. On these small exact simulators, subspace discovery and refresh cost more operator applications than direct full-space optimization.

```
cd code
python experiments/run_experiment.py \
  --config experiments/configs/maxcut_small.yaml
python experiments/summarize_results.py
```

The experiment runner writes one CSV row per method, instance, and seed. The summarizer reads only that CSV and regenerates the four figures, Table 1, and a machine-readable JSON summary. Tests cover the statevector against a dense reference, JVP/VJP consistency, fixed-tolerance factorization, spectral residual estimation, basis recycling, the forward-only zero-VJP invariant, baselines, and the optional PennyLane cross-check.

9 Conclusion

RSQ turns the local observable Jacobian of ma-QAOA into an adaptive optimization space without forming the Jacobian. Its certificate has a precise role: it bounds discarded first-order sensitivity and decides when the space must be updated. On the released benchmark, this mechanism preserves depth-one quality under substantial parameter compression, reveals a depth-two compression–quality tradeoff, and carries nontrivial setup cost. These findings establish a fact-checkable baseline for future randomized-subspace quantum optimization and define the evidence required before stronger query-efficiency claims are made.

Table 1: Released 36-instance benchmark. Approximation ratios are mean $\pm$ standard deviation. Paired changes are relative to full ma-QAOA on the same instance and are mean $\pm$ two standard errors in percentage points.

Method	Depth	Parameters	Approx. ratio	Paired change
Full ma-QAOA	1	22.4	0.851 ± 0.053	–
Symmetry-tied	1	11.6	0.822 ± 0.065	-2.92 ± 1.56 pp
Fixed-rank oracle	1	8.0	0.756 ± 0.047	-9.54 ± 2.18 pp
RSQ ($\varepsilon = 0.1$)	1	13.4	0.848 ± 0.054	-0.28 ± 0.86 pp
RSQ ($\varepsilon = 0.01$)	1	13.4	0.849 ± 0.053	-0.21 ± 0.89 pp
Forward-only RSQ	1	8.0	0.819 ± 0.049	-3.18 ± 1.43 pp
Full ma-QAOA	2	44.9	0.925 ± 0.057	–
Symmetry-tied	2	23.1	0.912 ± 0.072	-1.25 ± 1.37 pp
Fixed-rank oracle	2	8.0	0.760 ± 0.038	-16.55 ± 1.87 pp
RSQ ($\varepsilon = 0.1$)	2	13.4	0.897 ± 0.060	-2.80 ± 1.44 pp
RSQ ($\varepsilon = 0.01$)	2	13.4	0.896 ± 0.060	-2.86 ± 1.43 pp
Forward-only RSQ	2	8.0	0.829 ± 0.037	-9.62 ± 2.02 pp

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